

Inglés técnico - Nivel I

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Engineering

SCIENCE



WRITTEN BY: Ralph J. Smith

Engineering is the application of science to the optimum conversion of the resources of nature to the uses of humankind. The field has been defined by the Engineers Council for Professional Development, in the United States, as the creative application of “scientific principles to design or develop structures, machines, apparatus, or manufacturing processes, or works utilizing them singly or in combination; or to construct or operate the same with full cognizance of their design; or to forecast their behaviour under specific operating conditions; all as respects an intended function, economics of operation and safety to life and property.” The term engineering is sometimes more loosely defined, especially in Great Britain, as the manufacture or assembly of engines, machine tools, and machine parts.

The words engine and ingenious are derived from the same Latin root, *ingenerare*, which means “to create.” The early English verb *engine* meant “to contrive.” Thus, the engines of war were devices such as catapults, floating bridges, and assault towers; their designer was the “engine-er,” or military engineer. The counterpart of the military engineer was the civil engineer, who applied essentially the same knowledge and skills to designing buildings, streets, water supplies, sewage systems, and other projects.

Associated with engineering is a great body of special knowledge; preparation for professional practice involves extensive training in the application of that knowledge. Standards of engineering practice are maintained through the efforts of professional societies, usually organized on a national or regional basis, with each member acknowledging a responsibility to the public over and above responsibilities to his employer or to other members of his society.

The function of the scientist is to know, while that of the engineer is to do. The scientist adds to the store of verified, systematized knowledge of the physical world; the engineer brings this knowledge to bear on practical problems. Engineering is based principally on physics, chemistry, and mathematics and their extensions into materials science, solid and fluid mechanics, thermodynamics, transfer and rate processes, and systems analysis.

Unlike the scientist, the engineer is not free to select the problem that interests him; he must solve problems as they arise; his solution must satisfy conflicting requirements. Usually efficiency costs money; safety adds to complexity; improved performance increases weight. The engineering solution is the optimum solution, the end result that, taking many factors into account, is most desirable. It may be the most reliable within a given weight limit, the simplest that will satisfy certain safety requirements, or the most efficient for a given cost. In many engineering problems the social costs are significant.

Engineers employ two types of natural resources—materials and energy. Materials are useful because of their properties: their strength, ease of fabrication, lightness, or durability; their ability to insulate or conduct; their chemical, electrical, or acoustical properties. Important sources of energy include fossil fuels (coal, petroleum, gas), wind, sunlight, falling water, and nuclear fission. Since most

resources are limited, the engineer must concern himself with the continual development of new resources as well as the efficient utilization of existing ones.

Source: <https://www.britannica.com/technology/engineering>

a. Conteste las siguientes consignas en español:

1. Lea la primera oración de cada párrafo y piense en un subtítulo para cada uno de ellos.
2. ¿Cómo se define la ingeniería? Mencione la diferencia entre EEUU y Gran Bretaña.
3. ¿En qué disciplinas se basa la ingeniería?
4. ¿Qué recursos naturales son utilizados por los ingenieros? ¿De dónde proviene la energía?

b. Analice el origen de la palabra engineer. ¿Qué ocurrió con el concepto engine al agregarle el sufijo -er? Las terminaciones de las palabras (sufijos) nos ayudan a determinar su función en la oración: sustantivo, adjetivo, verbo, adverbio. Observe las palabras subrayadas en el texto y ubíquelas según su terminación en el siguiente cuadro. Tenga en cuenta el ejemplo dado.

SUSTANTIVO	ADJETIVO	ADVERBIO	VERBO
application			

c. Ubique en el texto las siguientes frases nominales (tenga en cuenta el contexto dado) y explique la función de la terminación -ing en cada una de ellas:

manufacturing processes

specific operating conditions

extensive training

floating bridges

engineering

designing buildings



Different Types of Metal Corrosion and Basic Preventative Coatings



It's common for high performance structural components to experience some form of corrosion regardless of what type of material is used. **Corrosion resistant coatings** can increase the lifespan of a part, as well as reduce maintenance and replacement costs, but in order to select the appropriate coating it's important to identify what kind of corrosion a part is prone to. Based on how a part is used and what conditions it's exposed to, the kind of corrosion that develops may differ. There are five general types of corrosion: **galvanic**, **stress cracking**, **general**, **localized** and **caustic agent corrosion**.

Galvanic corrosion is extraordinarily common, and occurs when two metals with different electrochemical charges are linked via a conductive path. Corrosion occurs when metal ions move from the anodized metal to the cathodic metal. In this case, a corrosion resistant coating would be applied to prevent either the transfer of ions or the condition that causes it. Galvanic corrosion can also occur when one impure metal is present. If a metal contains a combination of alloys that possess different charges, one of the metals can become corroded. The anodized metal is the weaker, less resistant one, and loses ions to the stronger, positively charged cathodic metal. Without exposure to an electrical current, the metal corrodes uniformly; this is then known as general corrosion.

Stress-corrosion cracking (SCC) can seriously damage a component beyond the point of repair. When subjected to extreme tensile stress, a metal component can experience SCC along the grain boundary—cracks form, which are then targets for further corrosion. There are multiple causes of SCC, including stress caused by cold work, welding, and thermal treatment. These factors, combined with exposure to an environment that often increases and intensifies stress-cracking, can mean a part goes from suffering minor stress-corrosion to experiencing failure or irreparable damage.

General corrosion occurs as a result of rust. When metal, specifically steel, is exposed to water, the surface is oxidized and a thin layer of rust appears. Like galvanic corrosion, general corrosion is also electrochemical. In order to prevent oxidation, a preventative coating must interfere with the reaction.

Localized corrosion occurs when a small part of a component experiences corrosion or comes in contact with specific corrosion-causing stresses. Because the small “local” area corrodes at a much fast rate than the rest of the component, and the corrosion works alongside other processes such as stress and fatigue, the end result is much worse than the result of stress or fatigue alone.

Caustic agent corrosion occurs when impure gas, liquids, or solids wear a material down. Although most impure gases do not damage metal in dry form, when exposed to moisture they dissolve to form harmful corrosive droplets. Hydrogen sulfide is an example of one such caustic agent.

Corrosion Resistant Coatings

Corrosion resistant coatings for metal vary depending on the kind metal involved and the kind of corrosion prevention needed. To prevent galvanic corrosion in iron and steel alloys, coatings made from zinc and aluminum are helpful. Large components, such as bridges and energy windmills, are

often treated with zinc and aluminum corrosion resistant coatings because they provide reliable long-term corrosion prevention. Steel and iron fasteners, threaded fasteners, and bolts are often coated with a thin layer of cadmium, which helps block hydrogen absorption which can lead to stress cracking.

In addition to cadmium, zinc, and aluminum coatings, nickel-chromium and cobalt-chromium are often used as corrosive coatings because of their low level of porosity. They are extremely moisture resistant and therefore help inhibit the development of rust and the eventual deterioration of metal. Oxide ceramics and ceramic metal mixes are examples of coatings that are strongly wear resistant, in addition to being corrosion resistant.

Other Chemicals Guides

- General Industrial Paint Components
- Hardwood Floors: Cleaning and Maintenance
- The History of Green Chemistry and Processes

A. Observe la organización del texto, sus componentes textuales o lingüísticos y paratextuales y conteste:

¿Cuál es el tema del texto?

¿Qué tipo de texto es? ¿Cuál es el propósito del autor?

B. Lea el texto y conteste:

1. ¿Cuál es el problema planteado en el primer párrafo? ¿Qué solución se ofrece?
2. ¿Qué se necesita para seleccionar el revestimiento adecuado?
3. ¿Cuántos tipos de corrosión existen y en qué consisten? Explique la corrosión general.

C. Busque los siguientes conectores en el texto y explique qué expresan:

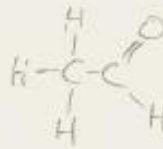
Conector	Párrafo en el texto	Expresa....	Significa...
As well as	1		
Either....or	2		
If	2		
Such as	7		
In addition to	8		
Therefore	8		

TEXT 3 -ENGINEERING CAREER



ENGINEERING YOUR FUTURE

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$$2\text{AgNO}_3 + \text{Fe} \rightarrow \text{Fe}(\text{NO}_3)_2 + 2\text{Ag}\downarrow$$


WHAT IS ENGINEERING?



ENGINEERS BRING DREAMS TO LIFE

ENGINEERS TAKE IDEAS AND TURN THEM INTO REALITY, USING SCIENCE, MATHS AND IMAGINATION. ENGINEERS ARE MASTERS OF PROBLEM-SOLVING AND CREATIVE DESIGN.

As a career, engineering offers a chance to make a difference to people's lives, while you enjoy a dynamic and varied working life.

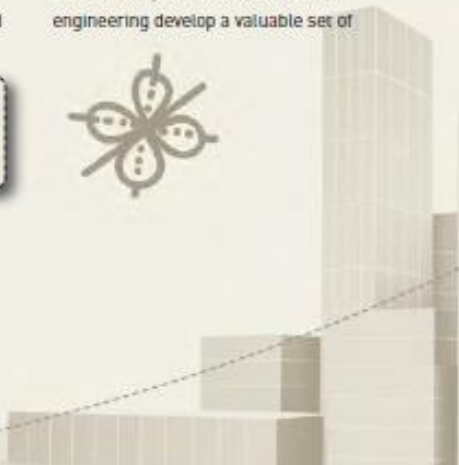
Engineers are in demand worldwide because they are essential to the growth and development of every country.

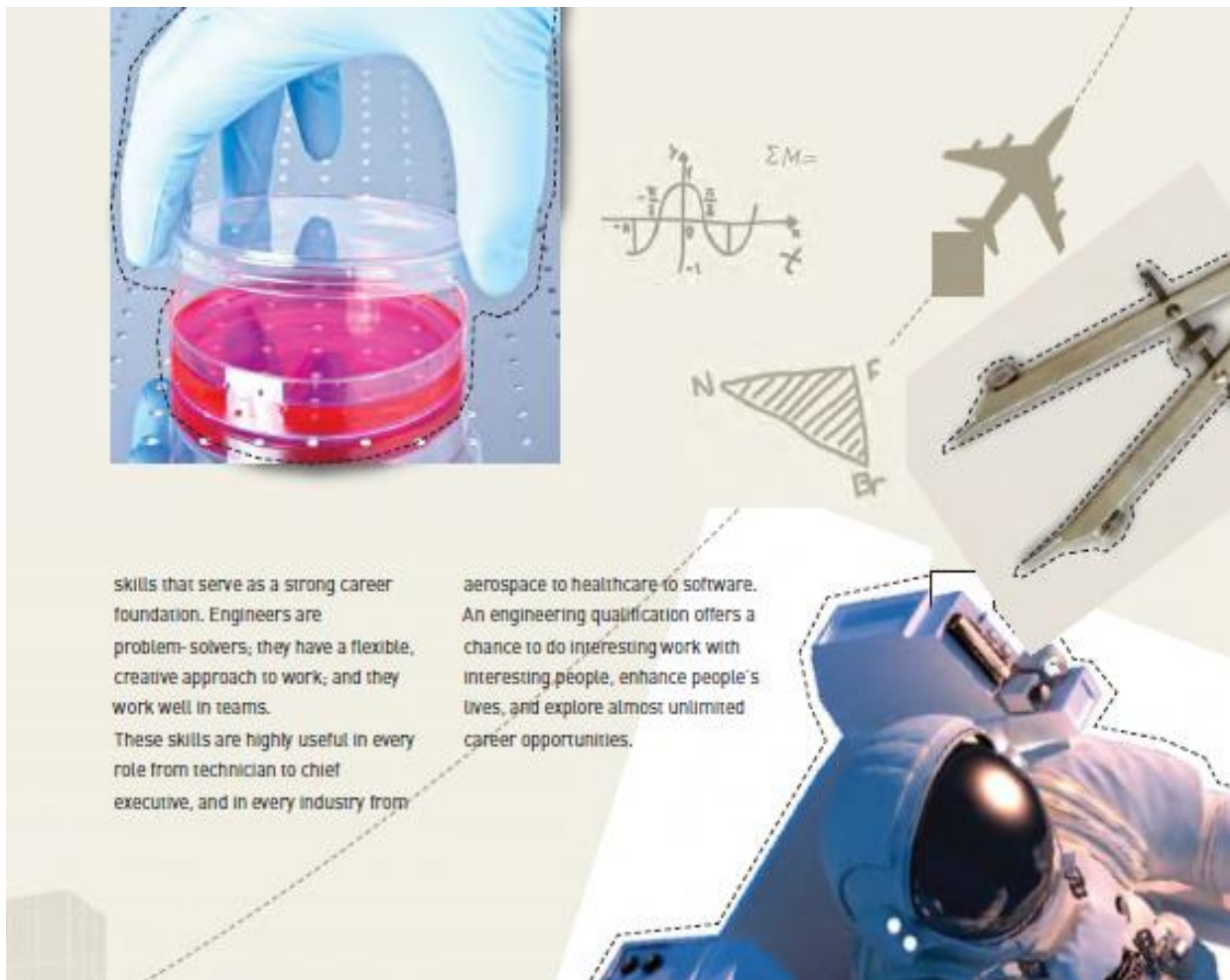
The progress of civilisation has largely depended on the work of engineers, who have harnessed and controlled our environment to provide us with essentials for living like water, shelter and power. In the past few decades, engineers have changed the world by helping to send

astronauts into space, mapping the human genome, and creating the internet.

Right now, engineers are working on innovative projects from solar energy to synthetic organs, driverless cars to virtual reality headsets. Students of engineering develop a valuable set of

This guide, brought to you by the Engineers Ireland STEPS programme, aims to answer the most commonly asked questions about engineering. See the back of this book for details of the services offered by the STEPS programme.





Responda las siguientes consignas:

1. ¿A qué género textual pertenece este artículo?
2. ¿Cuál es su fuente de publicación?
3. ¿A quién está dirigido este artículo?
4. ¿Observa una definición de ingeniería? ¿Puede realizar una comparación con el Texto 1?
5. Señale en el texto las palabras que terminan en -ING. Explique la función que tienen de acuerdo al contexto.

TEXT 4 - CLIMATE CHANGE

The screenshot shows the top navigation bar of the EPA website with the logo and text 'United States Environmental Protection Agency'. Navigation links include 'LEARN THE ISSUES', 'SCIENCE & TECHNOLOGY', 'LAWS & REGULATIONS', and 'ABOUT EPA'. There is an 'Advanced Search' field and an 'A-Z Index' link. Below the navigation bar, the 'Climate Change' section is highlighted, with a breadcrumb trail: 'You are here: EPA Home » Climate Change » Basic Information'. The main heading is 'Climate Change: Basic Information'. On the right, there are links for 'En Español', 'Contact Us', and 'Share'.

Climate change is happening

Our Earth is warming. Earth's average temperature has risen by 1.4°F over the past century, and is projected to rise another 2 to 11.5°F over the next hundred years. Small changes in the average temperature of the planet can translate to large and potentially dangerous shifts in climate and weather.

The evidence is clear. Rising global temperatures have been accompanied by changes in weather and climate. Many places have seen changes in rainfall, resulting in more floods, droughts, or intense rain, as well as more frequent and severe heat waves. The planet's oceans and glaciers have also experienced some big changes - oceans are warming and becoming more acidic, ice caps are melting, and sea levels are rising. As these and other changes become more pronounced in the coming decades, they will likely present challenges to our society and our environment.

What are climate change and global warming?

Global warming refers to the recent and ongoing rise in global average temperature near Earth's surface. It is caused mostly by increasing concentrations of greenhouse gases in the atmosphere. Global warming is causing climate patterns to change. However, global warming itself represents only one aspect of climate change.

Climate change refers to any significant change in the measures of climate lasting for an extended period of time. In other words, climate change includes major changes in temperature, precipitation, or wind patterns, among other effects, that occur over several decades or longer.

Earth's temperature is a balancing act

Earth's temperature depends on the balance between energy entering and leaving the planet's system. When incoming energy from the sun is absorbed by the Earth system, Earth warms. When the sun's energy is reflected back into space, Earth avoids warming. When energy is released back into space, Earth cools. Many factors, both natural and human, can cause changes in Earth's energy balance.

Scientists have pieced together a picture of Earth's climate, dating back hundreds of thousands of years, by analyzing a number of indirect measures of climate such as ice cores, tree rings, glacier lengths, pollen remains, and ocean sediments, and by studying changes in Earth's orbit around the

sun. ^[1]The historical record shows that the climate system varies naturally over a wide range of time scales. In general, climate changes prior to the Industrial Revolution in the 1700s can be explained by natural causes, such as changes in solar energy, volcanic eruptions, and natural changes in greenhouse gas (GHG) concentrations. ^[1]

Recent climate changes, however, cannot be explained by natural causes alone, especially warming since the mid-20th century. Rather, human activities can very likely explain most of that warming. Human activities release large amounts of carbon dioxide and other greenhouse gases into the atmosphere. The majority of greenhouse gases come from burning fossil fuels to produce energy, although deforestation, industrial processes, and some agricultural practices also emit gases into the atmosphere.

The Greenhouse Effect causes the atmosphere to retain heat

When sunlight reaches Earth's surface, it can either be reflected back into space or absorbed by Earth. Once absorbed, the planet releases some of the energy back into the atmosphere as heat (also called infrared radiation). Greenhouse gases (GHGs) like water vapor (H₂O), carbon dioxide (CO₂), and methane (CH₄) absorb energy, slowing or preventing the loss of heat to space. In this way, GHGs act like a blanket, trapping energy in the atmosphere and causing it to warm. This process is commonly known as the "greenhouse effect". It is natural and necessary to support life on Earth. However, the buildup of greenhouse gases can change Earth's climate and result in dangerous effects to human health and welfare and to ecosystems.

The Main Greenhouse Gases

The most important GHGs directly emitted by humans include CO₂, CH₄, nitrous oxide (N₂O), and several others. The sources are detailed below.

Carbon dioxide

Carbon dioxide is the primary greenhouse gas that is contributing to recent climate change. CO₂ is absorbed and emitted naturally as part of the carbon cycle, through animal and plant respiration, volcanic eruptions, and ocean-atmosphere exchange. Human activities, such as the burning of fossil fuels and changes in land use, release large amounts of carbon to the atmosphere, causing CO₂ concentrations in the atmosphere to rise.

Methane

Methane is produced through both natural and human activities. For example, natural wetlands, agricultural activities, and fossil fuel extraction and transport all emit CH₄. Methane is more abundant in Earth's atmosphere now than at any time in at least the past 650,000 years. ^[2] Due to human activities, CH₄ concentrations increased sharply during most of the 20th century.

Nitrous oxide

Nitrous oxide is produced through natural and human activities, mainly through agricultural activities and natural biological processes. Fuel burning and some other processes also create N₂O. Concentrations of N₂O have risen approximately 18% since the start of the Industrial Revolution, with a relatively rapid increase towards the end of the 20th century.

Other Greenhouse Gases

Water vapor is the most abundant greenhouse gas and also the most important in terms of its contribution to the natural greenhouse effect, despite having a short atmospheric lifetime. Some human activities can influence local water vapor levels. However, on a global scale, the concentration of water vapor is controlled by temperature, which influences overall rates of evaporation and precipitation.^[1]Therefore, the global concentration of water vapor is not substantially affected by direct human emissions.

Tropospheric ozone (O₃), which also has a short atmospheric lifetime, is a potent greenhouse gas. Chemical reactions create ozone from emissions of nitrogen oxides and volatile organic compounds from automobiles, power plants, and other industrial and commercial sources in the presence of sunlight. In addition to trapping heat, ozone is a pollutant that can cause respiratory health problems and damage crops and ecosystems.

Chlorofluorocarbons (CFCs), hydrochlorofluorocarbons (HCFCs), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), and sulfur hexafluoride (SF₆), together called F-gases, are often used in coolants, foaming agents, fire extinguishers, solvents, pesticides, and aerosol propellants. Unlike water vapor and ozone, these F-gases have a long atmospheric lifetime, and some of these emissions will affect the climate for many decades or centuries.

Other Climate Forcers

Black carbon (BC) is a solid particle or aerosol, not a gas, but it also contributes to warming of the atmosphere. Unlike GHGs, BC can directly absorb incoming and reflected sunlight in addition to absorbing infrared radiation. BC can also deposit on and darken snow and ice, increasing the snow's absorption of sunlight and accelerating melt.

Sulfates, organic carbon, and other aerosols can cause cooling by reflecting sunlight. Warming and cooling aerosols can interact with clouds, changing a number of cloud attributes such as their formation, dissipation, reflectivity, and precipitation rates. Clouds can contribute both to cooling, by reflecting sunlight, and warming, by trapping outgoing heat.

Lea el texto y responda en español:

1. ¿Cómo se manifiesta el cambio climático?
2. ¿Cuál es la diferencia entre cambio climático y calentamiento global?
3. Explique el proceso del cual depende la temperatura de la tierra.
4. ¿Qué factores pueden explicar el cambio climático?
5. ¿Por qué se produce el efecto invernadero?
6. Busque en el 2do párrafo un conector de adición y mencione los dos conceptos que une.
7. Otros conectores para analizar: -However - Therefore - In addition to
8. Subraye ideas principales y realice una síntesis en no más de 4 (cuatro) oraciones.



Tesla's Early Years

Nikola Tesla was born a subject of the Austro-Hungarian Empire in 1856 in a mountainous area of the Balkan Peninsula known as Lika. His father Milutin, and his mother Djuka, were both Serbian by origin. Tesla's father was a stern but loving Orthodox priest, who was also a gifted writer and poet. At a young age, Tesla immersed himself in his father's library. Tesla's mother was a hard-working woman of many talents who created appliances to help with home and farm responsibilities. One of these was a mechanical eggbeater. Tesla attributed all of his inventive instincts to his mother.

Tesla began his education at home and later attended gymnasium in Carlstadt, Croatia excelling in his studies along the way. An early sign of his genius, he was able to perform integral calculus in his mind, prompting his teachers to think he was cheating. During this period young "Niko" saw a steel engraving of Niagara Falls. In his imagination there appeared a huge water wheel being turned by the powerful cataract. He said to an uncle that he would go to America one day and capture energy in this way. Thirty years later he did exactly that. Despite his early creativity, Tesla did not begin to think of himself as an inventor until he was a young adult.

Passionate about mathematics and sciences, Tesla had his heart set on becoming an engineer but was "constantly oppressed" by his father's insistence that he enter the priesthood. At age seventeen, Tesla contracted cholera and craftily exacted an important concession from his father: the older Tesla promised his son that if he survived, he would be allowed to attend the renowned Austrian Polytechnic School at Graz to study engineering. Tesla's wish became a reality.

At the Polytechnic school Tesla began his studies in mechanical and electrical engineering. One day a physics teacher showed Tesla's class a new Gramme dynamo that—by employing direct current—could be used as both a motor and a generator. After watching it for a time, Tesla suggested it might be possible to do away with a set of inefficient sparking connections known as commutators. This, his amused professor said, would be like building a perpetual motion machine! Not even Tesla could hope to achieve such a feat. For the next several years the challenge obsessed Tesla, who instinctively knew that the solution lay in electric currents that alternated.

It wasn't until age twenty-four, when Tesla was living in Budapest and working for the Central Telephone Exchange, that the answer came to him:

One afternoon, which is ever present in my recollection, I was enjoying a walk with my friend in the city park and the idea came like a flash of lightning and in an instant the truth was revealed. I drew with a stick on the sand the diagram shown six years later in my address before the American Institute of Electrical Engineers.

This was the invention of the induction motor, a technological advance that would soon change the world.

Coming to America

Following his discovery in Budapest, Tesla was hired by electric power companies in Strasbourg and Paris to improve their DC generation facilities. In Germany and France, he attempted to interest investors in his concept for an AC motor but had no success. It was clear that in order to realize his idea, he would have to meet the greatest electrical engineer in the world—Thomas Alva Edison.

At age 28, Nikola Tesla arrived in New York City and was shocked by what he discovered. "What I had left was beautiful, artistic and fascinating in every way; what I saw here was machined, rough and unattractive. It [America] is a century behind Europe in civilization." The Serbian immigrant had four cents in his pocket, some mathematical computations, a drawing of an idea for a flying machine, and a letter of introduction from Charles Batchelor, one of Edison's business associates in Europe.

Electricity was first introduced to New York in the late 1870s. Edison's incandescent lamp had created an astonishing demand for electric power. And his DC power station on Pearl Street in lower Manhattan was quickly becoming a monopoly. On the streets, single poles carried dozens of crooked crossbeams supporting sagging wires, and the exposed electrical wiring was a constant danger. Unsuspecting children would scale the poles only to meet an untimely electrical demise. The residents of Brooklyn became so accustomed to dodging shocks from electric trolley tracks that their baseball team was called the *Brooklyn Dodgers*. In spite of the perils, wealthy New Yorkers rushed to have their homes wired, the most important being the banker, J.P. Morgan, who had invested heavily in Edison.

It was into this state of affairs that the 6'4" immigrant from Eastern Europe entered Edison's office. Thrilled and terrified to meet his hero, Tesla handed Edison his letter of recommendation: It read: "My Dear Edison: I know two great men and you are one of them. The other is this young man!" Tesla proceeded to describe the engineering work he had done, and his plans for an alternating current motor.

Edison knew little of alternating current and did not care to learn more about it. In short, AC power sounded like competition to Edison. But there was something different about Tesla, and Edison immediately hired him to make improvements in his DC generation plants. Tesla claimed that Edison promised him \$50,000 if he succeeded, perhaps thinking it an impossible undertaking. But the potential of so much money appealed mightily to the impoverished immigrant.

Both Tesla and Edison shared a common trait of genius in that neither of them seemed to need much sleep. Edison could go for days, taking occasional catnaps on a sofa in his office. Tesla claimed that his working hours at the Edison Machine Works were 10:30 a.m. till 5 a.m. the next day. Even into old age Tesla said he only slept two or three hours a night.

That's where the similarity ended. Tesla relied on moments of inspiration, perceiving the invention in his brain in precise detail before moving to the construction stage. Edison was a trial and error man who described invention as five percent inspiration and 95 percent perspiration. Edison was self-taught. Tesla had a formal European education.

It was only a matter of time until their differences would lead to conflict.

Several months after Edison employed him, Tesla announced that his work was successfully completed. When Tesla asked to be paid, however, Edison seemed astonished. He explained that the offer of \$50,000 had been made in jest. "When you become a full-fledged American you will appreciate an American joke," Edison said. Shocked and disgusted, Tesla immediately resigned.

Word began to spread that a foreigner of unusual talent was digging ditches to stay alive. Investors approached Tesla and asked him to develop an improved method for arc lighting. Although this was not the opportunity he had hoped for, the group was willing to finance the Tesla Electric Light Company. The proud new owner set to work and invented a unique arc lamp of beautiful design and efficiency. Unfortunately, all of the money earned went to the investors and all Tesla got was a stack of worthless stock certificates.

But his luck was about to change. Mr. A.K. Brown of the Western Union Company, agreed to invest in Tesla's idea for an AC motor. In a small laboratory just a short distance from Edison's office, Tesla quickly developed all the components for the system of AC power generation and transmission that is used universally throughout the world today. "The motors I build there," said Tesla, "were exactly as I imagined them. I made no attempt to improve the design, but merely reproduced the pictures as they appeared to my vision and the operation was always as I expected." The battle to produce his motor was over. But the struggle to introduce it commercially was only just beginning.

Source: <https://www.pbs.org/tesla/index.html>

Prelectura

- De acuerdo al título y subtítulos, ¿qué tipo de texto va a leer?
- ¿Tiene la organización textual canónica para este tipo de género textual? ¿En qué se basa para esta decisión?

Lectura

1. Explique con sus palabras por qué el equipo de baseball de Brooklyn, EEUU, se llama *Brooklyn Dodgers*.
2. ¿A qué o quién hacen referencia las frases "*The Serbian immigrant*" y "*the 6'4" immigrant from Eastern Europe*"? (Recuadrados en párrafos 8 y 10)
3. El autor expresa "That's where the similarity ended" (subrayado en el texto). ¿A qué similitud se refiere? ¿Entre quiénes? ¿Por qué llegó a su fin?
4. ¿Qué relación entre ideas expresa el conector *IF* resaltado en el texto? ¿Qué ideas conecta?
5. Ingrese a la página web sobre Tesla, Sección **Inside the Lab**, elija con su compañero una de las invenciones y sintetícela para el resto de sus compañeros.

TEXT 6 - VOLKSWAGEN BEETLE

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Volkswagen to end production of the iconic Beetle next year

[Rob Crilly](#), NEW YORK

14 SEPTEMBER 2018 • 1:51AM

Volkswagen is finally ending production of its iconic Beetle, ending an 80-year reign for a car that began life under Hitler's Third Reich, became a mainstay of popular culture through the 1960s and 1970s before being revived with an updated model in 1997.

The [German car maker](#) said the last vehicle would be made in Mexico next year, at the only factory still manufacturing the car. The company said the decision would allow **it** to focus on other models, including its portfolio of electric cars. Hinrich Woebcken, chief executive of Volkswagen Group of America, said: "The loss of the Beetle after three generations, over nearly seven decades, will evoke a host of emotions from the Beetle's many devoted fans." At its peak **it** sold more than a million cars a day, powered in part by the cinema exploits of Herbie the Love Bug.



Herbie in the 1968 film The Love Bug

But its utilitarian design – developed by Ferdinand Porsche to principles proposed by Adolf Hitler who wanted a functional, people's car or "volkswagen" – saw sales decline through the 1980s, fuelled in part by the rise of the VW Golf. Even so, its distinctive lines and unpretentious image meant it remained a counterculture favourite.

[Production of the original Beetle ended in 2003.](#) A sleeker version, which was bigger and featured more mod cons with whimsical touches (such as a dashboard vase) was introduced in 1997, after an experimental, one-off concept car version attracted huge excitement. It featured a water-cooled engine in the front – unlike the original which used an air-cooled system at the rear – and has been built at VW's factory in Puebla, Mexico, since 1999.

Sales dipped in recent years and efforts to [straighten its curves and add satnav](#), making it more appealing to men, only briefly stemmed the slide. VW sold 29,000 in 2012 but only 8627 last year, according to Autodata Corp.

A 2013 cabriolet

Analysts say it was out of sync with consumers in the US – the Beetle's biggest market – where bigger cars, including crossovers and sports utility vehicles, dominate the market. **However**, the company is not ruling out one day reviving such an instantly recognisable brand. "I would say 'never say never'," said Mr. Woebcken. Production will end with two special editions, featuring extra chrome, new wheels and three-colour ambient lighting inside.



--

Lea y responda las siguientes consignas en español:

Observe el texto, sus componentes lingüísticos y paralingüísticos:

1. ¿Qué tema trata?
2. ¿Qué tipo textual es? Descriptivo / argumentativo / publicitario / expositivo / narrativo / instructivo
3. ¿Cuál es la fecha y fuente de publicación?
4. ¿Qué información conoce sobre este tema?

Realice una primera lectura y conteste las siguientes consignas:

5. ¿Qué información brinda el texto sobre este Volkswagen desde la década del 60 a la actualidad?
6. ¿A qué se debió el auge y luego el declive en la producción de este automóvil?
7. ¿Cuál es la referencia del pronombre **it** marcado en negrita en el 2do párrafo?
8. En el último párrafo se observa el conector **However** (marcado en negrita). ¿Qué contraste observa el autor?
9. Mencione el propósito del autor al escribir este artículo.

Revolutionary Rail: High-Speed Rail Plan Will Bring Fast Trains to the U.S.

The next wave of high-speed rail lines should do away with the rails altogether, say proponents of magnetic levitation technology

By Stuart F. Brown | May 4, 2010 |



FAST LANE: California's planned high-speed line, shown here in an artist's impression, will likely be the first true high-speed line to be built in the U.S. With more than \$11 billion in financing secured, construction could begin as early as next year. *Image: COURTESY OF CALIFORNIA HIGH-SPEED RAIL AUTHORITY*

America is an absurdly backward country when it comes to passenger trains. As anyone who has visited Europe, Japan or Shanghai knows, trains that travel at nearly 200 miles per hour have become integral to the economies of many countries. With its celebrated Tokaido Shinkansen bullet trains, Central Japan Railway has for the past five decades carried billions of passengers between Tokyo and Osaka in half the time it would take to fly. A new Madrid-to-Barcelona express train runs at an average speed of 150 miles per hour; since its inception two years ago, airline traffic between the two cities has dropped by 40 percent. In contrast, Amtrak's showcase Acela train connecting Boston to Washington, D.C., averages just 70 mph. That figure is so low because many sections of the Acela's tracks cannot safely support high speeds, even though the train itself is capable of sprints above 150 mph. Think of it as a Ferrari sputtering down a rutted country lane.

There has been a recent push to change all this. Earlier this year the Department of Transportation announced the recipients of \$8 billion in stimulus funding designed to spread high-speed rail across the U.S. The 2010 federal budget requests an additional \$1 billion in rail construction funds in each of the next five years. And in 2008 California voters approved a \$9-billion bond measure to initiate an ambitious high-speed rail network that would connect Los Angeles to San Francisco and, eventually, Sacramento and San Diego.

Levitation System

Electronically controlled support magnets located on both sides along the entire length of the vehicle pull the vehicle up to the ferromagnetic stator packs mounted to the underside of the guideway.

Guidance magnets located on both sides along the entire length of the vehicle keep the vehicle laterally on the track. Electronic systems guarantee that the clearance remains constant (nominally 10 mm). To hover, the Transrapid requires less power than its air conditioning equipment. The levitation system is supplied from on-board batteries and thus independent of the propulsion system. The vehicle is capable of hovering up to one hour without external energy. While travelling, the on-board batteries are recharged by linear generators integrated into the support magnets.



1. Complete el siguiente cuadro:

Título	Autor
Fuente	Fecha de publicación

2. Contextualice los siguientes conceptos:

-Europe, Japan or Shanghai - Tokyo and Osaka - Los Angeles to San Francisco

3. Conteste las siguientes preguntas:

- ¿Por qué considera el autor que EEUU se encuentra en absurdo retroceso?
- ¿Por qué el tren que conecta Boston con Washington alcanza solo 70mph? ¿Con qué se lo compara? Explique la comparación.
- ¿Cuál fue la consecuencia de poner en marcha un tren express de Madrid a Barcelona?
- ¿Cuántos millones de dólares fueron solicitados en total en 2010 en EEUU para la mejora en el sistema de transporte ferroviario?

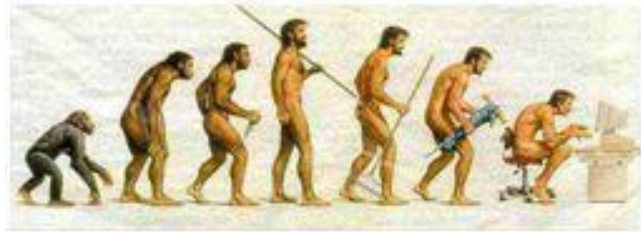
TEXT 8 - THE EVOLUTION OF TECHNOLOGY

THE BLOG

03/18/2010 05:12 am ET Updated Dec 06, 2017

The Evolution of Technology

By [Dr. Jim Taylor](#)



Have you ever thought about how far we've come in our ability to connect with others and how far we'll go? I've been thinking a lot about connectivity recently and have always found that looking back to where we came from can help us better understand where we are today and, more importantly, where we may be going in the future.

Consider the evolution of connectivity. We officially became homo sapiens approximately 200,000 years ago. With the emergence of sophisticated language, we were able to communicate with each other face to face in a way that allowed for the emergence of civilization as we now know it. Of course, this connection was limited to, well, shouting distance, but it was a start.

About 4,000 years ago, humans developed their first means of non-face-to-face communication with the discovery of smoke signals and then, about 2,500 years ago, drums. For the first time, people were able to connect without being in physical proximity to each other. Amazingly, not much changed in communication technology for the next 2,300 years or so.

Then, around 1835, Samuel Morse invented the telegraph, setting the stage for the greatest period of technological development in history that, in a relatively short time, has transformed our lives so dramatically. Think about it. The telegraph was a clear precursor to the Internet and the telegram was an early iteration of email.

Alexander Graham Bell's patent of the telephone in 1876 (many have laid claim to having invented it) enabled humans to converse directly over great distances as if they were in the same room.

The facsimile followed closely in the wake of the telephone, paving the way for the immediate transmission of something other than voice. For the first time, documents could be shared at a rate far faster than through the mail (what we now quaintly refer to as 'snail mail').

Mobile phone technology emerged for commercial use with the car phone around 1979 and progressively evolved to the present where mobile phones are now considered an indispensable part of our lives.

In 1994, the Internet was introduced to the public (it had actually been around since the 1960s), and it has likely been the single greatest leap forward in communication technology, enabling the instantaneous transmission of data, documents, still and moving images, and voice. It has created a veritable torrent of technology that has given us the Web, email, text messaging, and an array of applications, for example, MySpace, Facebook, Twitter, and Skype, that have dramatically altered the way we connect.

This brief and, admittedly, incomplete history provides a little perspective on how we arrived at the present. What did all of these communication technologies have in common? They have incrementally enabled us to connect with other people and access more information in more rapid, easy, and less costly ways. And each advancement changed our lives in ways manifest and subtle, direct and indirect, predictable and unexpected. Connectivity may be the most powerful tool in our lives today, with informational, economic, social, cultural, and political impact.

What then of the future of connectivity? What new technologies will be developed that will further change our lives? Perhaps we need look no further than science fiction to see what might become science fact in the not-too-distant future. Will we receive visual and auditory tweets through eyeglasses and ear pieces, respectively? Perhaps 3-D holographic telephone conversations? In the distant future, instead of voice recognition, how about thought recognition?

My concern is not in the technology itself; we cannot and should not try to slow or halt the inexorable march of progress. My interest is in our relationship with that technology and my concern is in how technology will affect us. Will we be passive recipients — dare I say victims? — of technology who allow it to change our lives for better or worse without consideration? Or can we be masters of our technology and deliberately harness its tremendous value while minimizing its risks?

The answer to these questions will depend not only on the technology itself that is developed, but also on our exploration of how new technology will influence our lives. Could anyone have predicted how the latest communication technology would change our lives? Maybe not, but I think it would be worth a try. Good questions to ask include:

1. What are our goals for this technology?
2. How will it influence how we interact with others?
3. How will it affect how we use our time?
4. What benefit will it bring to us?
5. What costs might arise from its use?
6. How can developers prepare us to best use this technology?

Yes, let us continue to nurture emerging technology to further connectivity. But the journey of progress shouldn't be guided by developers and engineers alone. Such a trip leaves behind other important aspects of connectivity, namely, our relationship with the technology itself,

where the risk is that the technology will lead us down a road of unintended consequences rather than our leading the technology down a road of our choosing.

Let's not forget that technology is not an end in itself, but rather a means to an end. What is that end? Enhancing the quality of our lives. Yet can we say unequivocally that the latest technology has done that? I'm not so sure. With that purpose in mind, bringing technologists together with those who reside at the nexus of technology and humanity, for example, experts from psychology, philosophy, and sociology, would be invaluable in answering these questions. Though computer and communication companies use neuroscientists in the "micro" development of technology (e.g., GUIs) and there is some academic study of these issues, I haven't found anything to indicate that technologists are exploring the "macro" side of technology (please correct me if I'm wrong).

Such a collaboration would serve two essential purposes. First, by fully understanding the relationship between technology and people, developers will actually create technology that will better serve our needs. Second, such a collaboration will increase the chances that we will understand the ramifications of new technology and ensure that it will provide the utmost benefit to humanity with only a minimum of costs.



Follow Dr. Jim Taylor on Twitter: www.twitter.com/DrJimTaylor

Dr. Jim Taylor Adjunct faculty, University of San Francisco

Lea y resuelva las siguientes consignas:

1. Complete el siguiente cuadro:

Título	
Autor	
Fecha de publicación	
Tipo textual	

2. ¿De qué trata el 1er párrafo? ¿Qué es la conectividad?
3. Realice una línea de tiempo en lo que refiere a tecnología leyendo solo la 1ra oración de cada párrafo, hasta que llegue al presente. ¿Hasta qué párrafo debe leer para completar la línea?
4. ¿Qué nuevas tecnologías serán desarrolladas que cambiarán nuestras vidas a futuro?
5. ¿A qué se refiere el autor cuando dice "my concern is not with the technology itself"? (Párr. 11)

6. ¿Qué frases nos dan la idea de un estilo informal en la redacción? Busque al menos 4 frases en el texto.
7. Lea los últimos tres párrafos y verifique si el autor contesta alguna de las preguntas mencionadas.
8. ¿Qué marcadores discursivos son importantes en el texto? ¿Por qué?

TEXT 9 - CLIMATE CHANGE WITH SUPER PLANTS

TOPICS

SEARCH

Los Angeles Times

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SCIENCE NOW SCIENCE LA TIMES

Scientists aim to fight climate change with super plants



By DEBORAH NETBURN NOV 17, 2017 | 1:25 PM



Ever since humans first walked the earth, we have relied on plants for our survival. They provide us with food, shelter, medicine and even the oxygen we breathe. Now, a team of scientists is wondering if they can protect us from climate change as well.

Researchers at the Salk Institute for Biological Studies in San Diego launched a new initiative to improve on the ability of plants to suck carbon dioxide out of the atmosphere and store it deep in the soil. They call it “Harnessing Plants.”

“There are a lot of geo-engineering efforts to come up with ways of pulling carbon dioxide out of the air,” said Joseph Noel, a chemical biologist at Salk who is working on the project. “Plants do this anyway, so why not try a biological solution as well.”

During the growing season, plants pull more than 100 gigatons of carbon out of the atmosphere through the process of photosynthesis. But much of that carbon is eventually released back into the air as CO₂ — either because we and other animals eat the plants or burn them, or they return to the soil where bacteria and fungi cause them to decompose.

The effects of this yearly cycle are measurable on a global scale. The concentration of carbon dioxide in the atmosphere consistently drops during the Northern Hemisphere's spring and summer, when plants are growing across the large land masses of North America, Europe and Asia. When winter descends and fewer plants are growing and others are decaying, the CO₂ concentrations rise once again.

One of the Salk team's goals is to find a way to help plants do a better job of taking the carbon they absorb from the atmosphere and keeping it in the soil.

All plants make a substance called suberin that protects their roots. It's the same material as the cork in your wine bottle or on your corkboard. It's also the material that makes up the skin of a potato.

The unique properties of suberin help plants in many ways, said Noel. It makes them more tolerant of drought and paradoxically, more tolerant of floods. Plants that grow in salt water produce a lot of suberin because it helps regulate how much salt is absorbed by their roots. It also serves as a protection against disease.

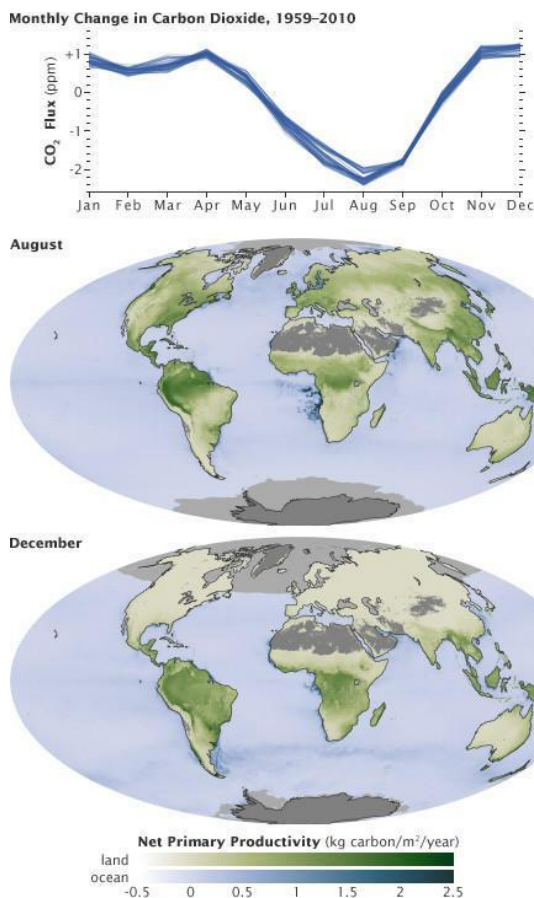
But perhaps most importantly for the group's goals, suberin is a carbon-rich polymer that is very difficult for bacteria and fungi to break down.

Further research in the lab revealed that suberin is one of the most stable forms of carbon in the soil. That means once carbon from the atmosphere makes it into the ground in the form of suberin, it will stay there.

Armed with this information, the group plans to breed a variety of plants that can produce more suberin than they currently do today. "We want them to make bigger roots and deeper roots with more suberin".

Of course, for their suberin-rich plants to have an impact on the global carbon cycle, they will have to be deployed on an enormous scale. In the longer term, the group envisions partnering with governments around the world to distribute seeds to farmers.

"We have to take as much as 1 trillion tons of carbon dioxide out of the air and as of now, there are no viable and scalable ways of taking carbon out of the air,"



Michael Strano, a chemical engineer who works with plants at MIT, noted that there are several advantages of using plants to sequester carbon. The only energy they need to do their work is harvested from the sun, plus they can regenerate themselves and are capable of self-repair.

Already the Salk Institute has invested more than \$7 million in the initiative, including building six high-tech climate control rooms that will allow the researchers to test seeds in a variety of climates, and future climates, from around the world.

Lea y responda las siguientes consignas.

a. Complete la siguiente información:

Tema del texto:

Fecha de publicación:

Fuente:

Género textual:

b. Responda las siguientes preguntas en español.

1. ¿En qué consiste el desarrollo descrito en el texto? ¿Qué evidencia se menciona para fundamentarlo?
2. ¿Qué es la *suberina*? Mencione sus propiedades.
3. ¿Cuál es la meta de los investigadores? ¿Cómo se relaciona la suberina con ésta?
4. ¿Tiene ventajas o desventajas utilizar plantas para el propósito de los investigadores? Justifique su respuesta.
5. ¿A qué se refiere THIS (subrayado en el texto)?

LIGHT READING: VISIBLE LIGHT COMMUNICATIONS

14 February 2011 | Updated: 14 February 2011 9:36 am | By [Ellie Zolfagharifard](#)

Advanced LED-based technology could revolutionise the way we send and receive data. Ellie Zolfagharifard reports.



The human desire for light was one of the reasons that caveman became engineer. From the very first camp fires to the oil-filled lamp and electric torch, man-made light has led the development of the modern world.

The objective has always been a simple one: to provide illumination in areas that the Sun can't reach. We've been extremely successful in this, with more and more places on Earth touched by artificial light each day. The technology has showcased our best innovations and given us the ability to see things that would otherwise be hidden.

But what if light could do more than just illuminate? What if it could also send streams of data? Traffic lights, television sets, car headlights, billboards and lamps might all suddenly become far more important in our daily lives. We could receive maps from a street light, get news alerts from lamps and download music from electronic posters.



On the move: VLC could be used in military operations where RF communications are restricted

It may sound like a futuristic concept, but a small community of researchers is already working on ways to make this a reality. They believe that an emerging area of technology known as Visible Light Communications (VLC), which uses the rapid flickering of advanced light-emitting diodes (LEDs) to encode data, could open up new and exciting possibilities in the way we send and receive information.

The work has been pioneered in Japan by the Visible Light Communications Consortium (VLCC). On the back of its research, the US has invested \$18.5m (£11.5m) in the development of VLC and the Chinese government is also thought to have put aside large sums to integrate it into aircraft. In Europe, Oxford and Edinburgh universities are involved in research, along with firms such as France Telecom and Siemens.

What has really excited researchers are the advantages of VLC over other forms of wireless communication. VLC doesn't interfere with radio-frequency (RF) electronics, making it suitable for use in hospitals and aircraft, and it has no associated health concerns. It's also environmentally friendly, with its use of existing infrastructure reducing costs and allowing for the future expansion of the network.

Prof Zabih Ghassemlooy, associate dean for research at Northumbria University, believes that safety is a major driver. 'We're now seeing people develop allergies to radiation from radio waves. Perhaps because of this, society will be reluctant to use them - light under moderate power is the way forward. Nature has provided us with this for billions of years and we should be making the most of it.'



Into the deep: VLC has promise for subsea communications

The main catalyst behind VLC has been improvements in the performance of LEDs - which are durable, efficient and bright, and are fast replacing incandescent lamps. Unlike normal light bulbs, LEDs respond quickly to 'on' and 'off' signals. By flickering an LED light on and off in a specific pattern, data can be sent at speeds that are undetectable to the human eye.

The most basic form of white LEDs is made up of a bluish to ultraviolet LED surrounded by a yellow phosphor, which emits white light when stimulated.

On average, these LEDs can achieve data rates of up to 40Mb/sec. Newer forms of LEDs, known as RGBs (red, green and blue), have three separate LEDs that, when lit at the same time, emit a light that is perceived to be white. As these involve no delay in stimulating a phosphor, data rates in RGBs can reach up to 100Mb/sec.

But it doesn't stop there. Resonant-cavity LEDs (RCLEDs), which are similar to RGB LEDs and are fitted with reflectors for spectral clarity, can now work at even higher frequencies. Last year, Siemens and Berlin's Heinrich Hertz Institute achieved a data-transfer rate of 500Mb/sec with a white LED, beating their earlier record of 200Mb/sec. As LED technology improves with each year, VLC is coming closer to reality and engineers are now turning their attention to its potential applications.

'In my view, there are two basic areas of application for VLC,' said Dr Dominic O'Brien from Oxford University. 'There is what you might call the "augmenting existing infrastructure" applications - using the solid-state lighting already present and adding a functionality - and applications where doing it in the visible region has an advantage in terms of security and performance.'

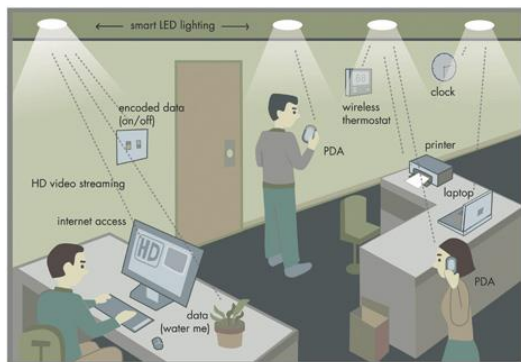
"Edison researched incandescent lamps and it changed the world – VLC will do the same"

One of the most promising applications is in car-to-car communication. If the headlights on a car could communicate with the tail lights of the car ahead, VLC collision-avoidance technology would be hugely significant in the automotive industry. In the same way, traffic lights could send detailed information of congestion up ahead directly to a vehicle. But, to be

successful, VLC has to prove itself against competing technologies of lidar, radar and RF, as well as to overcome some of its own technical challenges.

'The problem with using VLC outdoors is dealing with atmospheric conditions,' explained Ghassemlooy. 'Fog, smoke and temperature variation are major difficulties. We're looking at efficient modulation and coding schemes to see how we can push the beam through the fog without increasing power to the light source... I think we're getting closer to getting a solution every day.'

As well as problems with weather, VLC needs line-of-sight access to send data, restricting areas where it can operate. But O'Brien points out that its directional approach can also prove to be an asset. VLC is far more secure than RF signals, which move in many directions and can easily be intercepted. In military operations where RF-based communications are restricted during troop movements, VLC could be a viable alternative. For instance, it could be used to help securely pass information down a convoy of tanks and other military vehicles.



Ray of light: the US company claims that the LVX system saves up to 80 per cent on costs

But it's not just on the road that the technology holds promise. The directional nature of VLC makes it ideal for use in underwater communication systems. For instance, Canadian robotics research firm Penguin Automated Systems has developed a system made up of a spherical dome covered in LEDs, each switching on and off 40 million times a second. 'We've created a high-speed network that will control swarms of robots for large-scale subsea mining applications,' said chairman Dr. Greg Baiden. 'But it's not just for underwater. If we want to do orbital

construction and mining on the Moon, we need to get the bandwidth, and this could allow us to do that.'

Elsewhere, engineers at Niigata University in Japan are looking at using the technology to develop a positioning system that gets data from light fixtures. Liu Xiaohan, who is helping to develop the system, believes it could be used in applications such as guiding visually impaired people through hospital hallways. 'If we use LED and image sensors as the receiver, we can reach an accuracy of less than 5cm,' he said. 'It is far more accurate than other location technologies, but the biggest problem is cost. Adapting existing light fixtures for VLC functionality will be a huge task.'

The potential for VLC is huge and researchers are coming close to overcoming many of its technical challenges. Ghassemlooy is confident that VLC will eventually be accepted as an integral part of our infrastructure. He confessed to being worried about opposition from manufacturers with a vested interest in RF, but added that even their support will be won once the benefits of VLC become apparent.

'Visible light is the story of the human,' said Xiaohan. 'It's the first thing we experience when we come into this world and it's natural for us to want more. I believe there will be more and more people getting involved in VLC research. When Thomas Edison did research into incandescent lamps, it changed the world - VLC will do the same.'

With the research gathering momentum, the technical challenges facing VLC are getting smaller by the day. But the excitement and activity surrounding it is only just beginning. If it proves successful, the world could be facing a much brighter future.

Read more: <http://www.theengineer.co.uk/in-depth/the-big-story/light-reading-visible-light-communications/1007419.article#ixzz3q4atZ6c7>

Responda las siguientes consignas:

- 1) Lea los dos primeros párrafos introductorios y explique el contraste con el tercer párrafo.
- 2) ¿Cómo se llama y en qué consiste la nueva área tecnológica que se está empezando a investigar? Explique en qué sectores se puede aplicar la nueva tecnología "VLC".
- 3) ¿Cuáles son las dificultades que se mencionan en el uso de esta tecnología?
- 4) ¿Qué ventajas presenta?
- 5) Describa la aplicación al sistema vial de la tecnología en cuestión.
- 6) ¿Qué frases a lo largo del texto expresan opinión?
- 7) Al comienzo de dos párrafos hay un conector de adición y otro de contraste, ¿qué ideas relacionan?

TEXT0 11- UNIVERSITY PHYSICS WITH MODERN PHYSICS - PART 1

University Physics with Modern Physics

Hugh D. Young, Roger A. Freedman

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